

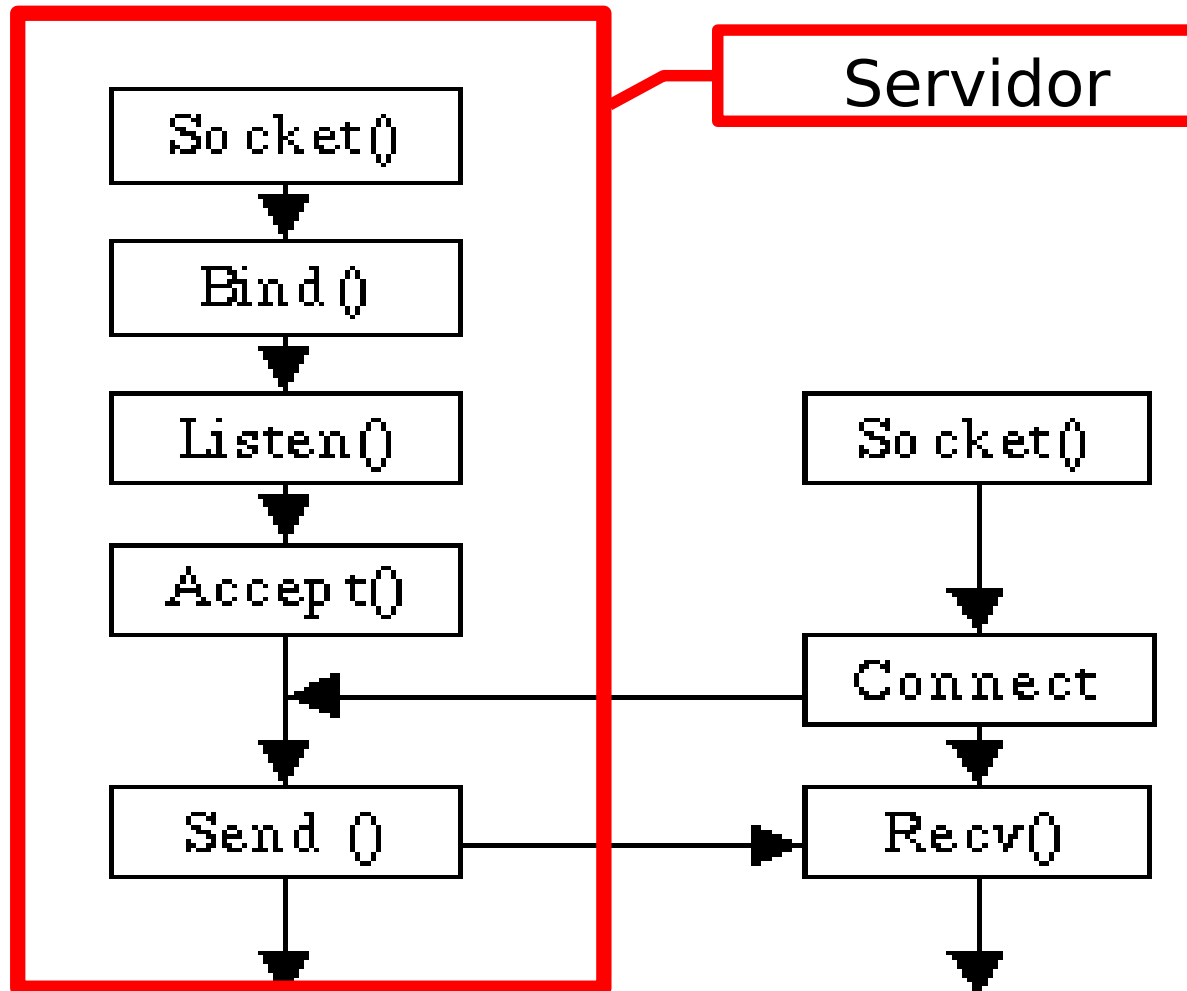
Sistemas Distribuídos

Winsock

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Programação com Sockets TCP/IP

Modelo TCP/IP



TCP/IP - Servidor

```
#include <winsock.h>
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#define N 50
```

```
int main(){
```

```
    char msg[N];
```

```
    int sock_srv, sock_cli;
```

```
    struct sockaddr_in addr_serv;
```

```
    int len;
```

```
    WSADATA ws;
```

```
    WSAStartup(0x0101,&ws);
```

```
    sock_srv = socket (AF_INET,SOCK_STREAM,0);
```

TCP/IP - Servidor

```
memset(&addr_serv,0, sizeof(addr_serv));
addr_serv.sin_family = AF_INET;
addr_serv.sin_port = htons(5000);
addr_serv.sin_addr.s_addr = htonl(INADDR_ANY);

bind(sock_srv,(struct sockaddr *)&addr_serv,sizeof(addr_serv));

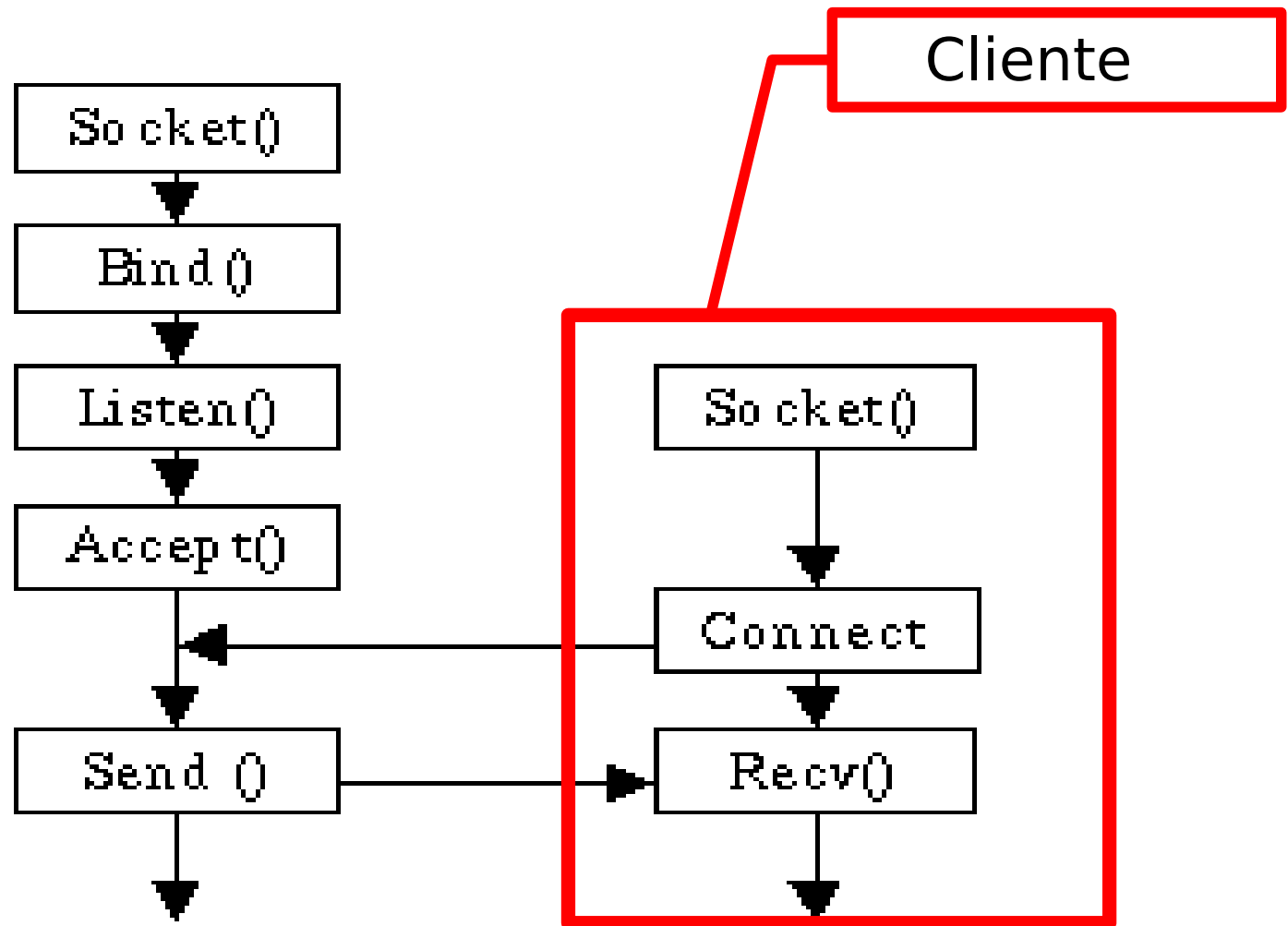
listen(sock_srv,3);

len = sizeof(addr_serv);
sock_cli = accept(sock_srv,(struct sockaddr *)&addr_serv, &len);

recv(sock_cli, msg, N, 0);
printf("Recebi: %s\n",msg);

close(sock_srv);
close(sock_cli);
return(0);
}
```

Modelo TCP/IP



TCP/IP - Cliente

```
#include <winsock.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define N 50

int main(){
    char msg[N];
    int sock;
    struct sockaddr_in addr;
    int len_addr;
    struct hostent *host;

    WSADATA ws;
    WSAStartup(0x0101,&ws);
```

TCP/IP - Cliente

```
sock = socket (AF_INET,SOCK_STREAM,0);

host = gethostbyname("localhost");
memset(&addr,0,sizeof(addr));
addr.sin_family = AF_INET;
addr.sin_port = htons(5000);
memcpy(&addr.sin_addr, host->h_addr,host->h_length);

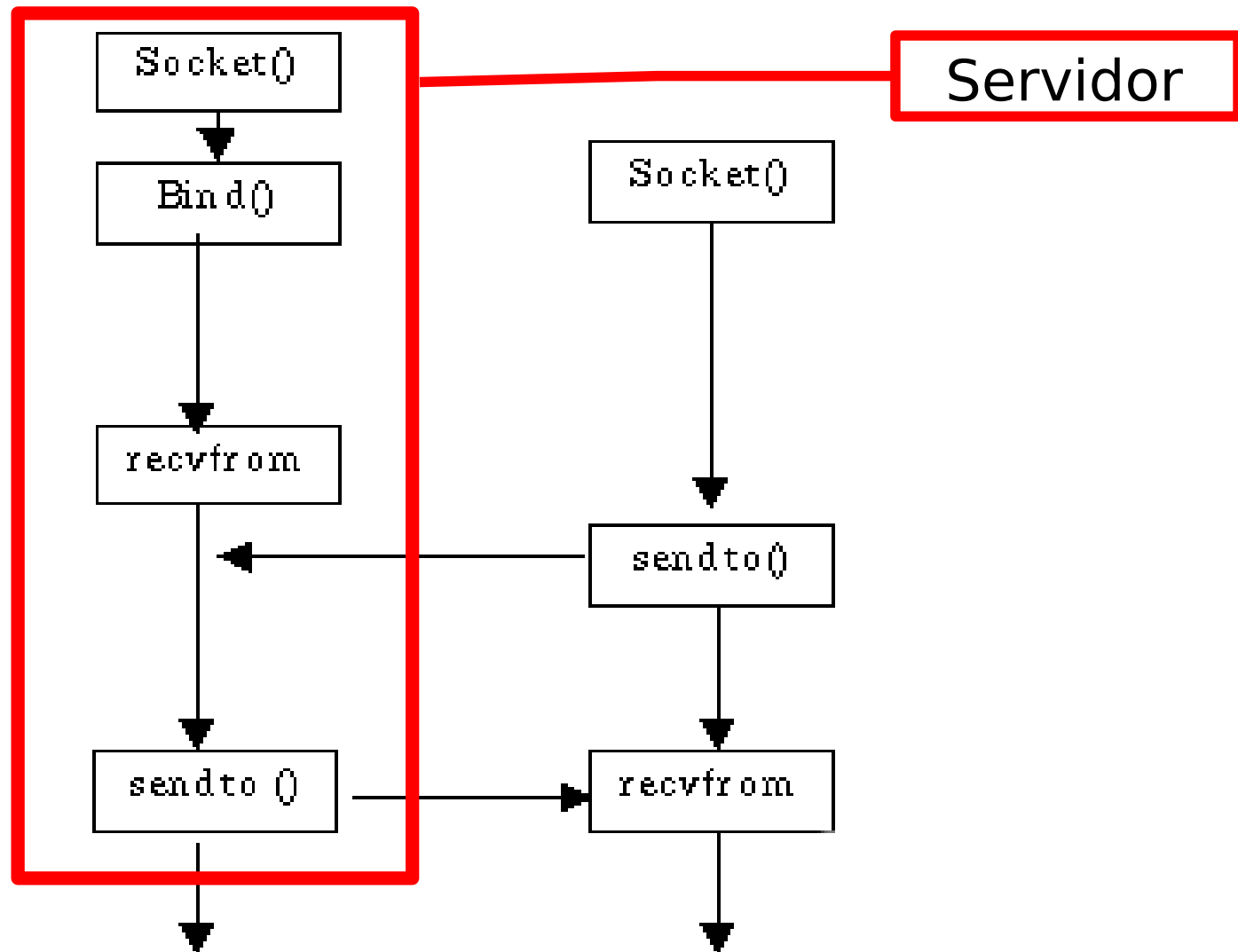
connect(sock, (struct sockaddr*)&addr, sizeof(addr));

printf("Digite a mensagem: ");
fgets(msg,50,stdin);

send(sock, msg, strlen(msg)+1, 0);
close(sock);
return(0);
}
```

Programação com Sockets UDP/IP

Modelo UDP/IP



UDP/IP - Servidor

```
#include <winsock.h>
```

```
#include <stdio.h>
```

```
#define N 50
```

```
int main(){
```

```
    SOCKET sock;
```

```
    struct sockaddr_in addr;
```

```
    int len;
```

```
    char msg[N];
```

```
    WSADATA ws;
```

```
    WSAStartup(0x0101,&ws);
```

```
    sock = socket(AF_INET, SOCK_DGRAM, 0);
```

UDP/IP - Servidor

```
addr.sin_family = AF_INET;  
addr.sin_port = htons(8000);  
addr.sin_addr.s_addr = htonl(INADDR_ANY);
```

```
bind(sock, (struct sockaddr *)&addr, sizeof(addr));
```

```
len = sizeof(addr);
```

```
recvfrom(sock, msg, N, 0, (struct sockaddr *)&addr, &len);
```

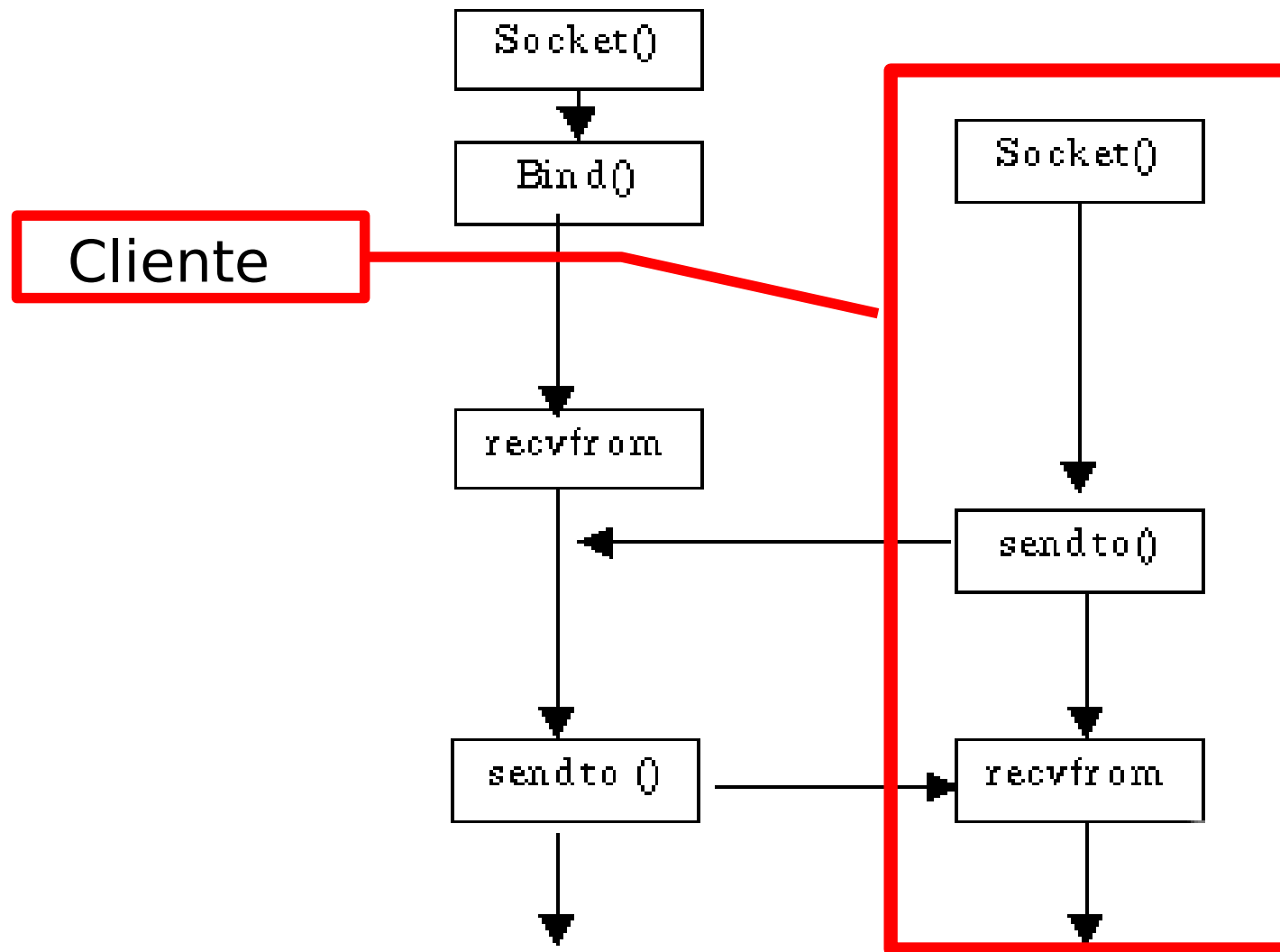
```
printf("Recebi: %s\n", msg);
```

```
closesocket(sock);
```

```
return 0;
```

```
}
```

Modelo UDP/IP



UDP/IP - Cliente

```
#include <winsock.h>
```

```
#include <stdlib.h>
```

```
#include <stdio.h>
```

```
#define N 50
```

```
int main(){  
    char msg[N];  
    SOCKET sock;  
    struct sockaddr_in addr;  
    int len;  
    struct hostent *host;
```

```
    WSADATA ws;
```

```
    WSAStartup(0x0101,&ws);
```

```
    sock = socket(AF_INET, SOCK_DGRAM, 0);
```

UDP/IP - Cliente

```
host = gethostbyname("localhost");
memset(&addr,0,sizeof(addr));
addr.sin_family = AF_INET;
addr.sin_port = htons(8000);
memcpy(&addr.sin_addr, host->h_addr,host->h_length);

printf("Digite a mensagem: ");
fgets(msg,50,stdin);

sendto(sock, msg, strlen(msg)+1, 0, (struct sockaddr
*)&addr, sizeof(addr));

close(socket(sock));

return(0);
}
```